

EXPERIMENT NO 19

Determination and plotting of Antenna Efficiency.

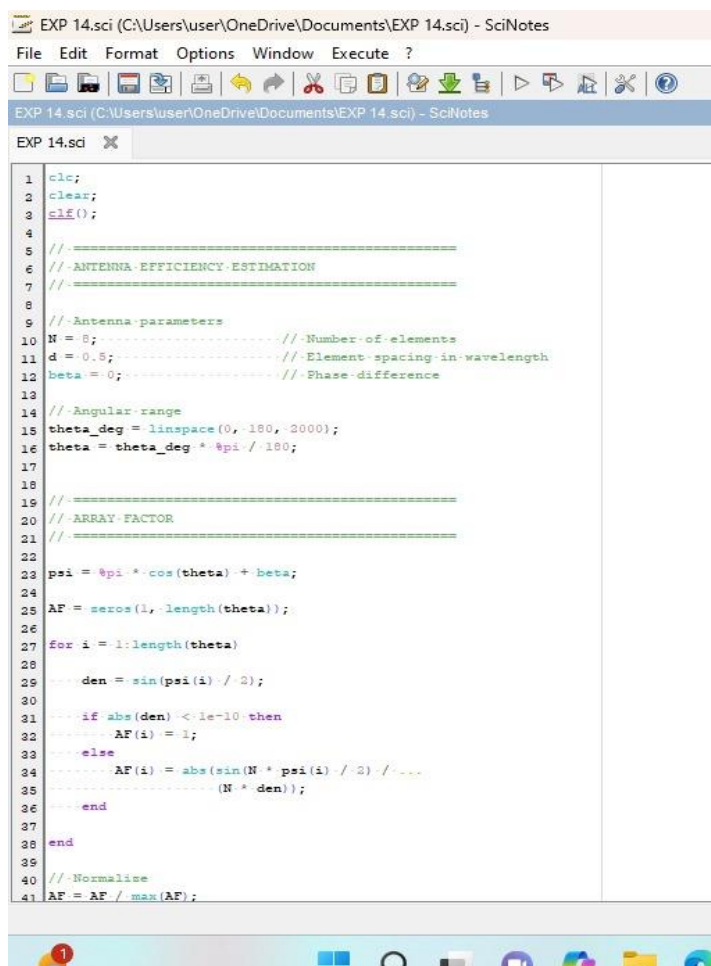
Aim:

To determine and plot Antenna Efficiency and to Estimate the efficiency based on radiated and input power.

Software required:

- SCILAB version 6.0.1

Program:



```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X
1  clc;
2  clear;
3  clf();
4
5  //=====
6  // ANTENNA EFFICIENCY ESTIMATION
7  //=====
8
9  //Antenna parameters
10 N = 8; // Number of elements
11 d = 0.5; // Element spacing in wavelength
12 beta = 0; // Phase difference
13
14 // Angular range
15 theta_deg = linspace(0, 180, 2000);
16 theta = theta_deg * pi / 180;
17
18
19 //=====
20 // ARRAY FACTOR
21 //=====
22
23 psi = pi * cos(theta) + beta;
24 AF = zeros(1, length(theta));
25
26 for i = 1:length(theta)
27
28     den = sin(psi(i) / 2);
29
30     if abs(den) < 1e-10 then
31         AF(i) = 1;
32     else
33         AF(i) = abs(sin(N * psi(i) / 2) / (N * den));
34     end
35 end
36
37
38 end
39
40 // Normalise
41 AF = AF / max(AF);
```

```
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File Edit Format Options Window Execute ?

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci x

121
122 plot(theta_deg, AF);
123
124 xlabel("Theta - (degrees)");
125 ylabel("Normalised Array Factor");
126 title("Array Factor vs Theta");
127
128 xgrid();
129
130 a = gca();
131 a.data_bounds = [0, 0, 180, 1.05];
132
133
134 // =====
135 // SAVE RESULTS TO TEXT FILE
136 // =====
137
138 filename = "antenna_results.txt";
139
140 fd = fopen(filename, "wt");
141
142 if fd == -1 then
143
144     disp("Error: Could not create output file.");
145
146 else
147
148     fprintf(fd, "ANTENNA PERFORMANCE RESULTS\n");
149     fprintf(fd, "=====\n");
150     fprintf(fd, "Number of Elements: %d\n", N);
151     fprintf(fd, "Element Spacing: %.2f lambda\n", d);
152     fprintf(fd, "Directivity: %.4f\n", directivity);
153     fprintf(fd, "Gain (dBi): %.2f\n", gain);
154     fprintf(fd, "Efficiency (%): %.2f\n", efficiency);
155
156     fclose(fd);
157
158     disp("Results saved to antenna_results.txt");
159
160 end
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Documents

- My Music
- My Pictures
- My Videos
- antenna_results.txt
- av 6.30
- EXP 14.sci

File/directory filter: Aa * x y

Scilab 2026.1.0 Console

```
=====
ANTENNA PERFORMANCE
=====
Number of Elements : 8
Element Spacing : 0.50 lambda
Directivity : 7.9997
Gain : 9.03 dBi
Efficiency : 25.00 %
=====

"Results saved to antenna_results.txt"

-->
```

Variable Browser

Name	Value	Type	Visibility	Memory
AF	1x2000	Double	local	16.2 kB
N	8	Double	local	216 B
P	1x2000	Double	local	16.2 kB
a	1x1 Graphic ...	Graphic	local	216 B
ans	0	Double	local	216 B
beta	0	Double	local	216 B
d	0.5	Double	local	216 B
den	-1	Double	local	216 B
directivity	8	Double	local	216 B

Command History

```
--> 6*10-6
6*10-7
4*10-7
-- 30/08/2026 11:26:37 --//
55
3*10-3
```

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

```
EXP 14.sci x
121
122 plot(theta_deg, AF);
123
124 xlabel("Theta - (degrees)");
125 ylabel("Normalised Array Factor");
126 title("Array Factor vs Theta");
127
128 xgrid();
129
130 a = gca();
131 a.data_bounds = [0, 0, 180, 1.05];
132
133
134 // =====
135 // SAVE RESULTS TO TEXT FILE
136 // =====
137
138 filename = "antenna_results.txt";
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140 fd = fopen(filename, "wt");
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149     fprintf(fd, "=====\n");
150     fprintf(fd, "Number of Elements: %d\n", N);
151     fprintf(fd, "Element Spacing: %.2f lambda\n", d);
152     fprintf(fd, "Directivity: %.4f\n", directivity);
153     fprintf(fd, "Gain (dBi): %.2f\n", gain);
154     fprintf(fd, "Efficiency (%): %.2f\n", efficiency);
155
156     fclose(fd);
157
158     disp("Results saved to antenna_results.txt");
159
160 end
```

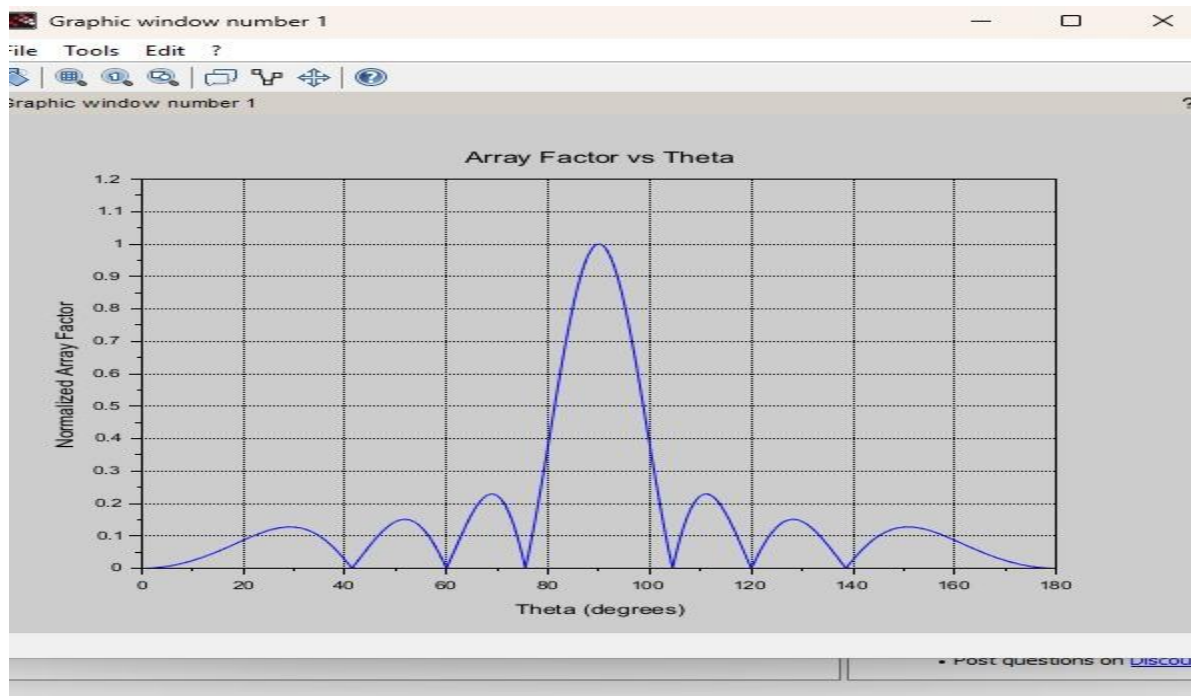
Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

12:29 PM 8/30/2026



Program:

```

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EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X

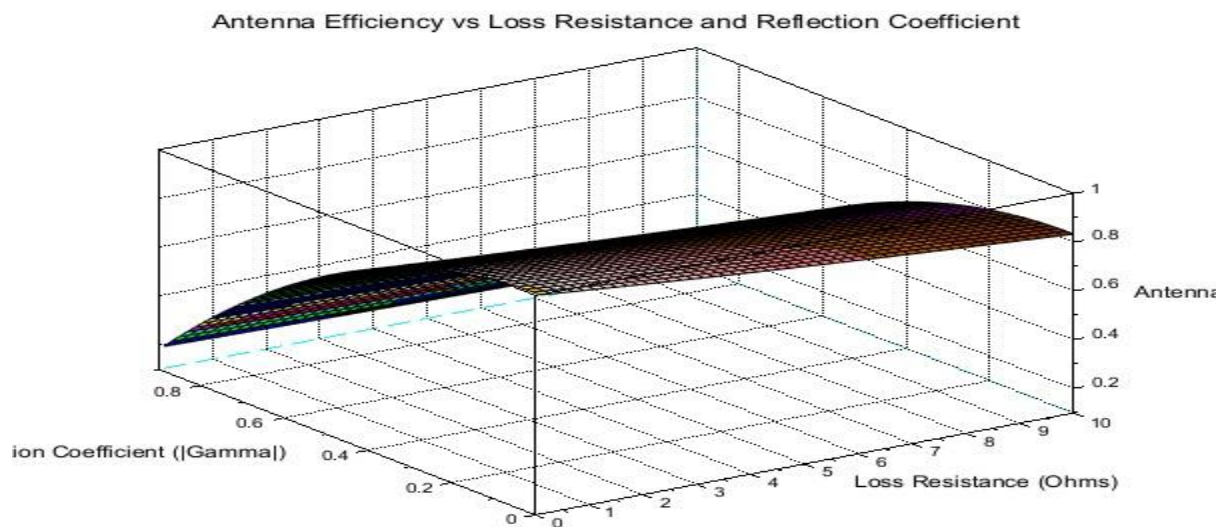
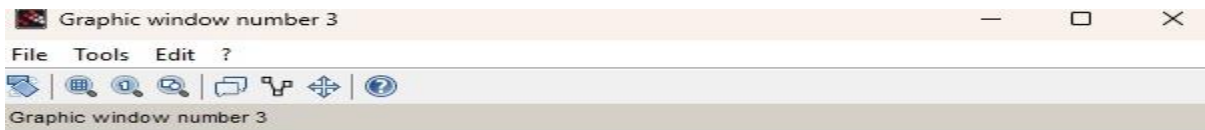
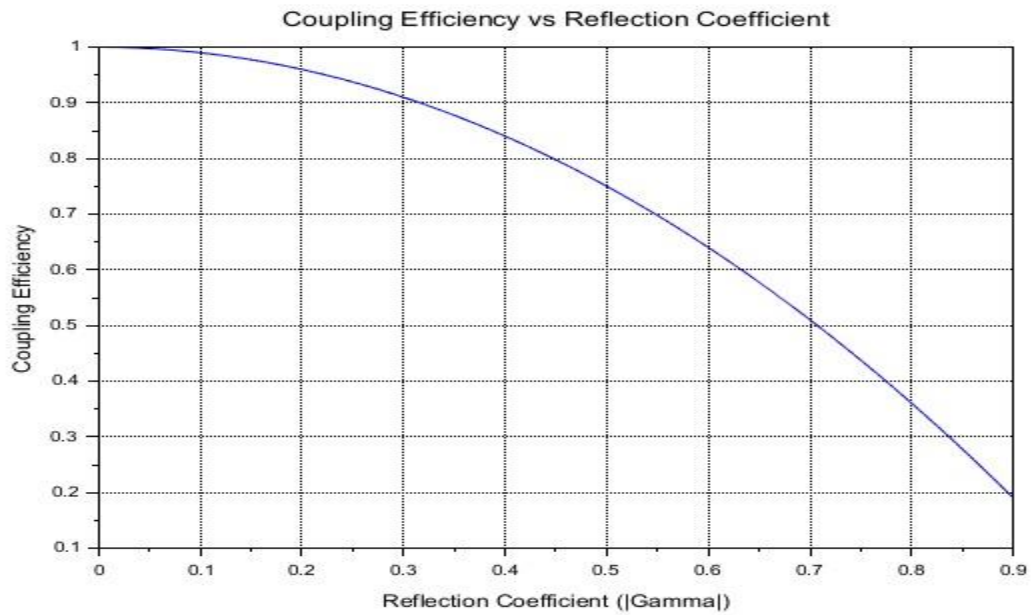
1  clc;
2  clear;
3  clf();
4
5  //=====
6  // ANTENNA EFFICIENCY ANALYSIS
7  //=====
8
9  // Loss resistance values in Ohms
10 R_l = linspace(0.1, 10, 50);
11
12 // Reflection coefficient magnitude
13 Gamma = linspace(0, 0.9, 50);
14
15 // Assume radiation resistance
16 R_rad = 50; % // Ohms
17
18 //=====
19 // RADIATION EFFICIENCY
20 // eta_r = R_rad ./ (R_rad + R_l)
21 //=====
22
23 e_r = R_rad ./ (R_rad + R_l);
24
25 //=====
26 // COUPLING EFFICIENCY
27 // eta_c = 1 - |Gamma|^2
28 //=====
29
30 e_c = 1 - Gamma.^2;
31
32 //=====
33 // ANTENNA EFFICIENCY MAP
34 // eta_a = eta_r * eta_c
35 //=====
36
37 // Create matrix for 2D map

```

```
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EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X
58 xlabel("Loss-Resistance- (Ohms)");
59 ylabel("Radiation-Efficiency");
60 title("Radiation-Efficiency-vs-Loss-Resistance");
61
62 xgrid();
63
64
65 // =====
66 // PLOT-2:-COUPLING-EFFICIENCY-VS-REFLECTION-COEFFICIENT
67 // =====
68
69 acf(2);
70
71 plot(Gamma, e_c);
72
73 xlabel("Reflection-Coefficient- ([Gamma])");
74 ylabel("Coupling-Efficiency");
75 title("Coupling-Efficiency-vs-Reflection-Coefficient");
76
77 xgrid();
78
79
80 // =====
81 // PLOT-3:- 2D-ANTENNA-EFFICIENCY-MAP
82 // =====
83
84 // -surf-requires-matrix-dimensions:
85 // -rows =-Gamma
86 // -columns =-R_l
87
88 acf(3);
89
90 surf(R_l, Gamma, e_a);
91
92 xlabel("Loss-Resistance- (Ohms)");
93 ylabel("Reflection-Coefficient- ([Gamma])");
94 xlabel("Antenna-Efficiency");
95 title("Antenna-Efficiency-vs-Loss-Resistance-and-Reflection-Coefficient");
96
97 xgrid();
98
```

Output:





RESULT:

The experiment was done and the efficiency of the given antenna was found. Different case studies were analyzed. These case studies highlight the practical importance of antenna efficiency in various applications. For all systems, increasing radiation efficiency (minimizing power loss) and coupling efficiency (ensuring impedance matching) were crucial for optimizing antenna performance. The calculations show that, for practical systems, small improvement